DSC-402, DSC-502 Group Project Phase 1

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Project: Data Visualization for IPEDS Distance Education Counts from their 12-Month Enrollment Survey

Students: Mason Turner, Ryan Anderson, Sean Syversen

Client: Rick Michalski, Director of Institutional Research and Effectiveness, UM-Dearborn

Data covered:

How the data will be used. The students will use the data only for this course project. It will be stored on the students' own devices or in a private repository, and shared only with the project team and the course instructor (for grading). The students will not share the data with anyone else, or use it for any other purpose, without the client's permission.

The client agrees to the following (check one in each group):

Who can see the finished visualizations

- ☑ Anyone (for example, on the student's public portfolio or Tableau Public)
- ☐ Classmates and the instructor only
- ☐ The instructor only

Whether the underlying data can be made public

- ☑ Yes, the data may be published with the visualizations
- ☐ Only in aggregated or anonymized form
- ☐ No, the data stays with the student(s)

After the course ends

- ☑ The student(s) may keep the data
- ☐ The student(s) will delete the data by (date)

Naming the client

- ☑ The client or organization may be named in the project
- ☐ The client should not be named

AI tools

- ☑ The data may be used with AI tools (for example, for help with code or analysis), acknowledging this may require sending data to a 3rd party server
- ☐ The data may only be used with local AI tools that allow all data to remain on the student(s) devices
- ☐ The data should not be entered into AI tools

Disclaimer: This agreement is not a legally binding contract. It was created for an educational course project at the University of Michigan–Flint and records, in plain language, what the students and client have agreed to. The students commit to following these terms in good faith.

Confirmed by

Client:

Rick Michalski rmichals@umich.edu 10/07/2026

Students:

Ryan Anderson, randrs@umich.edu 10/07/2026

Mason Turner, machtu@umich.edu 10/07/2026

Sean Syversen, ssyverse@umich.edu 10/12/2026

Requirements Brief

- **The ask and the real question:** General dashboard using the IPEDS data with options to filter the data with ease.

- **Audience:** Data and dashboard will be for public use. Primary consumer will be the UM-Dearborn Campus Leadership.

- **Purpose:** To provide an easy to view dashboard of the currently public data. Turn this raw data into something that can be viewed with ease.

- **Success statement:** Having access to a visualization that will allow me to assess online course delivery via the provided IPEDS data across institutions; with the ability to filter down to specific institutions and delineate between colleges and universities

- **Scope:** In scope is enrollment data, enrollment by course delivery type, enrollment by students from the Institution's state, out of state students, and students from outside of the United States. Out of scope for the project would be any data dictionary or referential materials.

- **Constraints:** No deadline constraint per client. Tools are all publicly available.

- **Open Questions:** Will the Tableau Dashboard continue to be hosted on the team's Github Page after the Semester is concluded?
    - No, client will need to merge files into his organizations environment and host from there.

Prioritized requirements: List requirements as must have, should have, could have, and won't have this time.

- Must have: Longitudinal Trends of total enrollment, enrolled in completely online, % enrolled completely online, % that reside outside of the United States.
- Should have: Ability to filter by College or University, ability to filter by location (e.g., state).
- Could have:
    - A parameter or switch that allows between total enrollment, enrolled in completely online, enrolled in some online, enrolled in none online.
    - A switch that changes the counts on the chart(s) from #'s to %'s (of total).
    - A parameter or switch that allows between % in state of institution, % out of state of the institution, % of out of US.
- Won't have this time
    - Additional IPEDS survey data not provided by client, that could be added in so users could view multiple surveys

How each will be checked. For each "must" and "should," say exactly how you will tell at delivery whether it was met. You can then use this list in your post-delivery evaluation at the end of the project.

- By confirmation with the client.

Stakeholders:

- UM-Dearborn Campus Leadership (e.g., Chancellor, Provost, Associate Provosts, Enrollment Management, Deans, etc.)
    - To more effectively access IPEDS data around longitudinal online course enrollments across the United States or by selected Regions or subgroups
- Other College and University Institutional Research Offices
    - To more effectively access IPEDS data around longitudinal online course enrollments across the United States or by selected Regions or subgroups
- *No one needs to approve use of this information as it is publicly available and the client intends to publish this dashboard on a public page and share broadly

Description of Data

The Data that was provided is sourced from the IPEDS Data Webpage. There are 3 datasets provided for the Project's requirements. The first dataset contains basic information about each educational institution including their name, location, unique identifier, etc. The second dataset contains the raw numbers of Fall Enrollment, and the number of students who are enrolled for in-person, online or mixed courses, as well as the number who are in-state, out of state, and international students. The third dataset is a dictionary about what each column name refers to.